

Assume contrarily that the equation has an integer solution (x_0, y_0) .

$$x_0^2 - 5y_0^2 = 3$$

$$\Rightarrow x_0^2 \equiv 5y_0^2 \equiv 2y_0^2 \pmod{3}$$

Notice that $\nexists z \in \mathbb{Z}$ such that $z^2 \equiv 2 \pmod{3}$. Thus, we have the following two cases:

$$(a) y_0^2 \equiv 0 \pmod{3}$$

$$(b) y_0^2 \equiv 1 \pmod{3}$$

However, note that (b) implies that $x_0^2 \equiv 2 \pmod{3}$ which is again impossible. Thus, (a) is the only possibility, implying that:

$$x_0^2 \equiv y_0^2 \equiv 0 \pmod{3} \Rightarrow 3 \mid x_0^2, 3 \mid y_0^2$$

$$\Rightarrow 3 \mid x_0, 3 \mid y_0$$

Thus, $x_0 = 3a$

$$y_0 = 3b$$

for some $a, b \in \mathbb{Z}$.

Thus, $x_0^2 - 5y_0^2 = 9(a^2 - 5b^2) = 3$

$$\Rightarrow a^2 - 5b^2 = \frac{1}{3}$$

which is impossible as $a, b \in \mathbb{Z}$. Thus, our assumption is incorrect & hence there does not exist any integer solution to the given equation $\square$ Q.E.D

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